SAT Math Trick Pack 2

84 new challenge items

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (20 items)

1. **A-316** · Hard · SPR · **No calculator needed**

The function f is linear, $f(3) = k$ and $f(k) = 3$, where k is a constant and $k \neq 3$. If $f(10) = -1$, what is the value of k?

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2. **A-317** · Hard · Multiple choice · **No calculator needed**

The function f is linear, $f(5) + f(-5) = 14$, and $f(2) = 13$. What is the value of $f(-2)$?

A) -13

B) 1

C) -1

D) 7

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3. **A-318** · Hard · SPR · **No calculator needed**

The function f is linear, $f(2) + f(8) = 30$, and $f(11) = 33$. What is the value of $f(0)$?

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4. **A-319** · Medium · Multiple choice · **No calculator needed**

In the xy-plane, what is the y-intercept of the graph of $x = 4y - 12$?

A) $(0, -12)$

B) $(0, -3)$

C) $(-12, 0)$

D) $(0, 3)$

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5. **A-320** · Hard · Multiple choice · **No calculator needed**

In the equation $ax + by = c$, a, b and c are constants with $a > 0$, $b < 0$ and $c > 0$. The graph of the equation in the xy-plane does NOT pass through which quadrant?

A) Quadrant II

B) Quadrant III

C) Quadrant I

D) Quadrant IV

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6. **A-321** · Hard · Multiple choice · **No calculator needed**

The system $2y - x > 4$ and $y < 5$ is graphed in the xy-plane. Which inequality describes all possible x-coordinates of points in the solution region?

A) $x > 6$

B) $x < 14$

C) $x < 6$

D) $x < -4$

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7. **A-322** · Hard · Multiple choice · **No calculator needed**

For real numbers x and y, $x + y < 12$ and $x - y > 4$. Which of the following must be true?

A) $y > 4$

B) $x > 8$

C) $y < 4$

D) $x < 8$

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8. **A-323** · Hard · SPR · **No calculator needed**

If $-3 \leq x \leq 2$ and $-5 \leq y \leq 1$, what is the greatest possible value of xy?

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9. **A-324** · Medium · Multiple choice · **No calculator needed**

A farm has 40 hectares of date palms and 25 hectares of olive trees. The equation $40x + 25y = 3{,}600$ represents the farm's total harvest, in tonnes, last season. What does x represent?

A) The average date harvest per hectare, in tonnes

B) The number of tonnes harvested per date palm tree

C) The number of hectares of date palms

D) The total date harvest, in tonnes

10. **A-325** · Medium · Multiple choice · **No calculator needed**

A line in the xy-plane has slope 3 and passes through the points $(2, k)$ and $(k, 14)$. What is the value of k?

A) 8

B) 4

C) 11

D) 5

11. **A-326** · Hard · SPR · **No calculator needed**

In the xy-plane, the region with $x \geq 0$, $y \geq 0$ and $2x + y \leq 12$ is a triangle. The line $y = mx$ divides this triangle into two regions of equal area. What is the value of m?

12. **A-327** · Medium · Multiple choice · **No calculator needed**

The function $V(t) = 1{,}200 - 0.75t$ gives the volume, in liters, of water in a tank t minutes after it starts draining. Which function gives the volume, in liters, h hours after the tank starts draining?

A) $1{,}200 - 0.75(h + 60)$

B) $1{,}200 - 45h$

C) $20 - 0.75h$

D) $1{,}200 - 0.0125h$

13. **A-328** · Hard · SPR · **No calculator needed**

In the system $y = 2x + k$ and $y = -x + 10$, k is a constant. For how many integer values of k is the solution (x, y) in Quadrant I (that is, $x > 0$ and $y > 0$)?

14. **A-329** · Medium · Multiple choice · **No calculator needed**

In the inequality $kx + 4 < 10$, k is a constant and $k < 0$. Which describes the solution set?

A) $x < \frac{6}{k}$

B) $x > \frac{6}{k}$

C) $x > -\frac{6}{k}$

D) $x < -\frac{6}{k}$

15. **A-330** · Medium · SPR · **No calculator needed**

A shopper spends exactly 600 dollars on x kilograms of oranges at 4 dollars per kilogram and y kilograms of apples at 5 dollars per kilogram, so $4x + 5y = 600$. If the shopper buys 10 more kilograms of oranges and still spends 600 dollars, by how many kilograms does y decrease?

16. **A-331** · Hard · Multiple choice · **No calculator needed**

If $7x - 2y = 3x + 6y$ and x and y are nonzero, what is the value of $\dfrac{x+y}{x-y}$?

A) The value cannot be determined from the information given.

B) $\dfrac{1}{3}$

C) 2

D) 3

17. **A-332** · Hard · Multiple choice · **No calculator needed**

Consider the system $3x - 4y = 10$ and $-6x + 8y = -20$. Which point lies on the graphs of both equations for every real number t?

A) $\left(t, \dfrac{3t-10}{4}\right)$

B) $\left(t, \dfrac{10-3t}{4}\right)$

C) $\left(t, \dfrac{3t+10}{4}\right)$

D) $\left(\dfrac{4t-10}{3}, t\right)$

18. **A-333** · Hard · Multiple choice · **No calculator needed**

A wall is $9x$ meters long, where x is a positive integer. Lamps are placed at both ends of the wall and every $\dfrac{9}{7}$ meters along it. Which expression gives the number of lamps?

A) $\dfrac{81x}{7} + 1$

B) $7x$

C) $7x + 1$

D) $9x + 1$

19. **A-334** · Medium · Multiple choice · **No calculator needed**

How many solutions does the system $2x + y = 11$, $x - y = 1$, $x + 3y = -13$ have?

A) Exactly one

B) Exactly three

C) Zero

D) Exactly two

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20. **A-335** · Hard · SPR · **No calculator needed**

An oven's temperature is $70°$F when it is switched on at $t = 0$. For $3 \leq t \leq 12$, where t is the time in minutes, the temperature is modeled by $T = a + bt$, where a and b are constants. The model gives $T = 200$ at $t = 3$ and $T = 380$ at $t = 7$. What is the value of a?

Advanced Math (19 items)

21. **AM-303** · Hard · Multiple choice · **No calculator needed**

In the equation $(k - 2)x^2 + 4x + 1 = 0$, k is a constant. The equation has exactly one real solution for exactly two values of k. What is the sum of these two values of k?

A) 8

B) 4

C) 2

D) 6

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22. **AM-304** · Hard · SPR · **No calculator needed**

The function g is defined by $g(x) = \dfrac{x^2 - kx - 18}{x + 3}$, where k is a constant. The graph of $y = g(x)$ in the xy-plane is a line with exactly one point removed. What is the y-coordinate of the removed point?

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23. **AM-305** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x) = \dfrac{x^2 - 4x - 12}{x^2 - 36}$. What is the sum of all values of x for which $h(x) = 0$?

A) 4

B) -8

C) 6

D) -2

24. **AM-306** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x) = x^2 - 9$ for $x < 2$ and $h(x) = -2x + 10$ for $x \geq 2$. What is the sum of all values of m for which $h(m) = 0$?

A) 0

B) 2

C) 5

D) 8

25. **AM-307** · Hard · Multiple choice · **No calculator needed**

In the equation $\frac{x^2}{x-a} = \frac{9}{x-a}$, a is a constant. For how many real values of a does the equation have exactly one real solution?

A) Infinitely many

B) Exactly two

C) Exactly one

D) Zero

26. **AM-308** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = x^2 - 1$. How many distinct real solutions does the equation $f(f(x)) = 0$ have?

A) 1

B) 4

C) 2

D) 3

27. **AM-309** · Hard · Multiple choice · **Desmos**

How many solutions (x, y) does the system $x^2 + y^2 = 25$ and $y = x^2 - 5$ have?

A) 3

B) 4

C) 1

D) 2

28. **AM-310** · Hard · Multiple choice · **Desmos**

How many real solutions does the equation $x^2 = 2^x$ have?

A) 1

B) 2

C) 3

D) 4

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29. **AM-311** · Medium · SPR · **No calculator needed**

What is the sum of all solutions to the equation $(x-3)^2 = (2x+1)^2$?

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30. **AM-312** · Hard · Multiple choice · **No calculator needed**

How many real solutions does the equation $\sqrt{x^2 - 10x + 25} = 5 - x$ have?

A) Exactly two

B) Zero

C) Infinitely many

D) Exactly one

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31. **AM-313** · Hard · Multiple choice · **No calculator needed**

A kiosk's daily profit, in hundreds of dollars, is modeled by $P(x) = -2x^2 + 12x + 5$, where x is the price of an item in dollars. The model is used only for $4 \le x \le 7$. What is the greatest daily profit, in hundreds of dollars, predicted by the model on this interval?

A) 21

B) 23

C) -9

D) 5

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32. **AM-314** · Hard · Multiple choice · **No calculator needed**

The expression $\dfrac{6x^2 + 7x - 5}{2x + 3}$ is equivalent to $Ax + B + \dfrac{C}{2x+3}$, where A, B and C are constants. What is the value of $A + B + C$?

A) -4

B) 2

C) 4

D) 0

33. **AM-315** · Hard · Multiple choice · **No calculator needed**

When the polynomial $p(x)$ is divided by $x - 5$, the remainder is 8. What is the remainder when $p(2x - 1)$ is divided by $x - 3$?

A) 15

B) 8

C) 5

D) 3

34. **AM-316** · Hard · Multiple choice · **No calculator needed**

For all real numbers x, $f(2x - 1) = 4x^2 + 3$. Which of the following defines $f(x)$?

A) $f(x) = x^2 + 4$

B) $f(x) = x^2 + 2x + 4$

C) $f(x) = 4x^2 + 3$

D) $f(x) = 4x^2 - 4x + 4$

35. **AM-317** · Hard · Multiple choice · **Desmos**

In the xy-plane, the graphs of $y = |x - 2|$ and $y = mx + 1$, where m is a constant, intersect at exactly two points. Which of the following could be the value of m?

A) -2

B) $-\frac{3}{4}$

C) 2

D) $\frac{1}{2}$

36. **AM-318** · Medium · SPR · **No calculator needed**

How many integers x satisfy the inequality $x(x - 4) < 2x$?

37. **AM-319** · Hard · SPR · **No calculator needed**

The function f is defined by $f(x) = a\big((x-3)^2 - b\big)\big((x-3)^2 - c\big)$, where a, b and c are constants. If $f(1) = 40$ and $f(6) = -14$, what is the value of $f(5) + f(0)$?

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38. **AM-320** · Hard · SPR · **No calculator needed**

In the equation $\dfrac{x+a}{x^2-a^2} = -\dfrac{3}{20x}$, a is a constant. If $x = 6$ is a solution, what is the value of a?

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39. **AM-321** · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{x^2}{\sqrt{x^2-k^2}} = \dfrac{k^2}{\sqrt{x^2-k^2}} + 24$, k is a positive constant. Which of the following is a solution?

A) $-\sqrt{k^2+576}$

B) $\sqrt{576-k^2}$

C) $\sqrt{k^2+24}$

D) $k+24$

Geometry and Trigonometry (26 items)

40. **GT-232** · Hard · Multiple choice · **No calculator needed**

In the xy-plane, a circle has equation $(x-4)^2 + (y+2)^2 = 50$. Which point lies in the interior of the circle?

A) $(10, 2)$

B) $(9, 3)$

C) $(-1, -6)$

D) $(-3, 0)$

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41. **GT-233** · Hard · SPR · **No calculator needed**

The graph of $x^2 + y^2 + 10x - 8y + k = 0$ in the xy-plane is a circle, where k is a constant. What is the greatest possible integer value of k?

42. GT-234 · Hard · Multiple choice · **No calculator needed**

For $0 < t < \frac{\pi}{2}$, $\sin t = 0.6$. What is the value of $\cos(\pi + t) + \sin(\pi - t)$?

A) 0.2

B) -1.4

C) -0.2

D) 1.4

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43. GT-235 · Hard · Multiple choice · **No calculator needed**

Points A, B and C lie on a circle with radius 12, and C lies on the major arc AB. The measure of angle ACB is $\frac{2\pi}{9}$ radians. What is the length of minor arc AB?

A) $\frac{16\pi}{3}$

B) $\frac{8\pi}{3}$

C) $\frac{64\pi}{3}$

D) $\frac{56\pi}{3}$

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44. GT-236 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that the measure of angle AED equals the measure of angle ABC. If $AD = 4$, $DB = 11$ and $AE = 6$, what is the length of $\overline{EC}$?

A) 10

B) 16.5

C) 22.5

D) 4

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45. GT-237 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle, $AC = 16$ and $BC = 12$. Point E lies on $\overline{AC}$ with $AE = 10$, and point F lies on $\overline{AB}$ so that $\overline{EF} \perp \overline{AB}$. What is the length of $\overline{EF}$?

A) 8

B) 6

C) 7.5

D) 12.5

46. **GT-238** · Medium · Multiple choice · **No calculator needed**

A triangle has two sides of lengths 10 and 12, and the angle between them measures $150°$. What is the area of the triangle?

A) $30\sqrt{3}$

B) 30

C) 15

D) 60

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47. **GT-239** · Hard · SPR · **No calculator needed**

A rhombus has side length 13 and area 120. What is the sum of the lengths of its diagonals?

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48. **GT-240** · Hard · Multiple choice · **No calculator needed**

A right circular cone stands with its vertex pointing up. Its base radius is 6 and its height is 12. Water fills the cone to a depth of 6, measured from the base. What is the volume of the water?

A) 18π

B) 72π

C) 108π

D) 126π

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49. **GT-241** · Hard · SPR · **No calculator needed**

A sector of a circle has radius 10 and central angle $216°$. The sector is rolled into a right circular cone so that its two straight edges meet. The volume of the cone is $k\pi$. What is the value of k?

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50. **GT-242** · Hard · Multiple choice · **No calculator needed**

Line ℓ passes through $(0, 0)$ and $(a, 12)$, where $a > 0$. The acute angle between ℓ and the positive y-axis measures $60°$. What is the value of a?

A) $12\sqrt{3}$

B) $4\sqrt{3}$

C) 24

D) 6

51. GT-243 · Hard · Multiple choice · **No calculator needed**

In triangles PQR and XYZ, $PQ = XY = 9$ and $m\angle R = m\angle Z = 35°$, where $\overline{PQ}$ is opposite angle R and $\overline{XY}$ is opposite angle Z. Which additional fact is enough to prove that triangle PQR is congruent to triangle XYZ?

A) $PR = XZ$

B) $QR = YZ = 14$

C) $m\angle Q = m\angle Y$

D) $PR = QR$

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52. GT-244 · Hard · Multiple choice · **No calculator needed**

A square has one side on the diameter of a semicircle of radius 10, and its other two vertices lie on the arc. What is the area of the square?

A) 100

B) 50

C) 80

D) 200

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53. GT-245 · Medium · Multiple choice · **No calculator needed**

In the xy-plane, the endpoints of a diameter of a circle are $(-2, 3)$ and $(6, -1)$. Which equation represents the circle?

A) $x^2 + y^2 - 4x - 2y - 15 = 0$

B) $x^2 + y^2 - 4x - 2y - 20 = 0$

C) $x^2 + y^2 - 4x - 2y - 75 = 0$

D) $x^2 + y^2 + 4x + 2y - 15 = 0$

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54. GT-246 · Hard · Multiple choice · **No calculator needed**

Triangle ABC has area 60. Point D lies on $\overline{BC}$ with $BD : DC = 2 : 3$, and point E lies on $\overline{AD}$ with $AE : ED = 3 : 1$. What is the area of triangle BED?

A) 15

B) 18

C) 10

D) 6

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55. GT-247 · Hard · Multiple choice · **No calculator needed**

In a circle of radius 13, two parallel chords have lengths 24 and 10. Which of the following could be the distance between the two chords?

A) 12

B) 17

C) 26

D) 5

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56. GT-248 · Medium · Multiple choice · **Calculator**

The radius of a right circular cylinder is increased by 20% and its height is decreased by 25%. What is the effect on the volume of the cylinder?

A) It does not change.

B) It increases by 8%.

C) It decreases by 5%.

D) It increases by 19%.

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57. GT-249 · Hard · Multiple choice · **No calculator needed**

Sphere Q has volume 36π. Sphere P has a radius 7 times the radius of sphere Q. The spheres are externally tangent at point T. Segment $\overline{MT}$ is a diameter of sphere P and segment $\overline{TN}$ is a diameter of sphere Q, with M, T and N on one line. What is the length of $\overline{MN}$?

A) 66

B) 45

C) 24

D) 48

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58. GT-250 · Hard · SPR · **Desmos**

In the equation $\sin\left(\frac{39\pi}{p}\right) = \frac{\sqrt{3}}{2}$, p is a constant and $p > 60$. What is the value of p?

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59. GT-251 · Hard · Multiple choice · **No calculator needed**

A rectangle is inscribed in a circle with area 676π. The length of the rectangle's diagonal is twice the length of its shorter side. What is the perimeter of the rectangle?

A) $52 + 52\sqrt{3}$

B) 104

C) $26 + 26\sqrt{3}$

D) $104 + 52\sqrt{3}$

60. **GT-252** · Hard · SPR · **No calculator needed**

Two identical right rectangular prisms each have a square base with side length s and a height of 35. Each prism has surface area K. The prisms are glued together base to base, and the new solid has surface area $\frac{15}{8}K$. What is the value of s?

61. **GT-253** · Hard · Multiple choice · **Calculator**

Square pyramids X and Y have the same base, a square with side length 10. Pyramid X has height 12 and pyramid Y has height 24. The total surface area of Y is approximately what percent greater than the total surface area of X?

A) 89%

B) 100%

C) 64%

D) 50%

62. **GT-254** · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $18°$. Which additional fact is NOT sufficient to determine the measure of angle B?

A) $m\angle B = m\angle C$

B) $m\angle C = 3\,m\angle B$

C) Triangle ABC is isosceles.

D) $m\angle C = 90°$

63. **GT-255** · Hard · Multiple choice · **Calculator**

Model Earth as a sphere with radius $6{,}371$ kilometers. An observer's eye is 1 kilometer above the surface. The line of sight from the eye to the horizon is tangent to Earth. Which is closest to the distance, in kilometers, from the eye to the point where the line of sight touches Earth?

A) 112.9

B) 1.0

C) $6{,}371.0$

D) 79.8

64. GT-256 · Hard · Multiple choice · **No calculator needed**

Parallel lines m and n are cut by a transversal. Angles b and c are same-side interior angles, and angle a is one of the other angles formed. If $200 < a + b + c < 236$, where all measures are in degrees, which of the following could be the measure of angle a, in degrees?

A) 56

B) 18

C) 64

D) 40

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65. GT-257 · Hard · Multiple choice · **No calculator needed**

In triangle JKL, the measure of angle J is $(ab)^\circ$ and the measure of angle K is $(cd)^\circ$, where $ab + cd = 90$. If $\sin J = 0.6$, what is the value of $\tan K$?

A) $\frac{3}{4}$

B) $\frac{4}{3}$

C) $\frac{4}{5}$

D) The value cannot be determined from the information given.

Problem-Solving and Data Analysis (19 items)

66. PS-157 · Hard · Multiple choice · **No calculator needed**

Data sets A and B each contain 20 values, all from the list $10, 20, 30, 40, 50$. The frequencies in A are $2, 3, 10, 3, 2$ and the frequencies in B are $5, 3, 4, 3, 5$, in that order. Which statement is true?

A) The standard deviations are equal because the means and ranges are equal.

B) The standard deviation of B is greater.

C) The standard deviation of A is greater.

D) The standard deviations cannot be compared without calculating them.

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67. PS-158 · Medium · SPR · **No calculator needed**

The daily high temperatures in a city for one week have a mean of $34°$C and a standard deviation of $4°$C. Each temperature is converted to degrees Fahrenheit using $F = 1.8C + 32$. What is the standard deviation of the converted temperatures, in degrees Fahrenheit?

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68. **PS-159** · Hard · SPR · **No calculator needed**

The data set $12, 15, 19, 24, 30$ has a range of 18. A sixth value, x, is added, and the range of the new data set is 1.5 times the original range. What is the sum of all possible values of x?

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69. **PS-160** · Hard · SPR · **No calculator needed**

The data set $2, 6, 9, 15, x$ has a mean equal to its median, where x is a real number. What is the sum of all possible values of x?

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70. **PS-161** · Hard · SPR · **No calculator needed**

A data set contains the values $1, 2, 3, 4, 5$ with frequencies $4, k, 6, 3, 5$, respectively, where k is a positive integer. The median of the data set is 3. What is the greatest possible value of k?

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71. **PS-162** · Hard · Multiple choice · **No calculator needed**

The commute times, in whole minutes, of 20 students are grouped: 10 students took 0 to 9 minutes, 7 took 10 to 19 minutes, and 3 took 20 to 29 minutes. Which of the following could NOT be the median commute time?

A) 10

B) 7.5

C) 14

D) 4

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72. **PS-163** · Hard · Multiple choice · **No calculator needed**

Data set A has 11 values with median 40. Data set B has 11 values with median 52. The two data sets are combined. Which of the following could be the median of the combined 22 values?

A) 46

B) 38

C) 60

D) 54

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73. **PS-164** · Hard · Multiple choice · **Calculator**

Electric vehicles made up 4% of a country's new-car sales in one year and 6% the next year. Total new-car sales increased by 10% from the first year to the second. By what percent did the number of electric vehicles sold increase?

A) 2%

B) 60%

C) 65%

D) 50%

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74. **PS-165** · Hard · SPR · **No calculator needed**

The number a is 150% more than the number b, and b is 60% less than the positive number c. The number a is what percent of c?

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75. **PS-166** · Hard · Multiple choice · **Calculator**

District A has $120{,}000$ residents and a population density of $4{,}000$ residents per square kilometer. District B has $60{,}000$ residents and a population density of $1{,}500$ residents per square kilometer. Which is closest to the population density, in residents per square kilometer, of the two districts combined?

A) $5{,}500$

B) $3{,}167$

C) $2{,}571$

D) $2{,}750$

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76. **PS-167** · Hard · SPR · **Calculator**

Bottle B holds 20% more juice than bottle A and costs 15% more. To the nearest tenth of a percent, by what percent is the price per liter of bottle B less than the price per liter of bottle A?

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77. **PS-168** · Hard · SPR · **No calculator needed**

In a school, a student is chosen at random. The probability that the student is in grade 10, given that the student is in the robotics club, is $\frac{3}{8}$. The probability that the student is in the robotics club, given that the student is in grade 10, is $\frac{1}{4}$. There are 96 grade 10 students. How many robotics club members are NOT in grade 10?

78. **PS-169** · Hard · SPR · **No calculator needed**

In grade 10, 12 students are in a club and 28 are not. In grade 11, k students are in the club and 42 are not. Being in the club is independent of grade. What is the value of k?

79. **PS-170** · Hard · Multiple choice · **No calculator needed**

In grade 10, 18 students are in the robotics club and 42 are not. In grade 11, 30 students are in the robotics club and 60 are not. A student is chosen at random from the students who are in grade 10 OR in the robotics club. What is the probability that the student is in grade 11?

A) $\frac{1}{3}$

B) $\frac{1}{5}$

C) $\frac{1}{2}$

D) $\frac{5}{8}$

80. **PS-171** · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.5x + 40$. For the data point with $x = 24$, the predicted value of y is 20% less than the actual value. What is the residual for this data point?

81. **PS-172** · Hard · SPR · **No calculator needed**

The model $\hat{y} = 4.8x + 62$ estimates a town's population y, in thousands, where x is the number of decades since 1980. According to the model, what is the estimated increase in the population, in people, per year?

82. **PS-173** · Medium · SPR · **No calculator needed**

The ratio of grade 9 students to grade 10 students in a club is $3 : 5$, and the ratio of grade 10 students to grade 11 students is $4 : 7$. The club has 201 students in grades 9, 10 and 11. How many more grade 11 students than grade 9 students are in the club?

83. **PS-174** · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.4x + 15$. A new data set is formed by adding 5 to every x-value and keeping each y-value. What is the y-intercept of the line of best fit for the new data set?

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84. **PS-175** · Hard · SPR · **No calculator needed**

Data set A has 10 integers: 4 are from 10 to 19 and 6 are from 20 to 29. Data set B has 10 integers: 3 are from 30 to 39 and 7 are from 40 to 49. What is the smallest possible value of (mean of B) $-$ (mean of A)?
